

Koo Ho yin (Jack)

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🐙 GitHub

🌐 LinkedIn

UG CS & Phys & AI — HKUST

EDUCATION

- **Bachelor of Science**

2022–2026

Hong Kong University of Science and Technology

- **1st Major:** Computer Science and Engineering
- **Extended Major:** Artificial Intelligence
- **2nd Major:** Physics (Honors Option)
- **Minor:** Mathematics
- **CGA:** 3.82/4.3 **MCGA:** 3.90/4.3

- **Exchange Programs**

Nanyang Technological University (Singapore)

2025 Summer

Freie Universität Berlin (Germany)

2026 Winter

PROJECTS

- **AI Weather Forecaster Chatbot**

2026 Jan–July

Website Developer Intern, RAG Application

Hong Kong Observatory

- Built a RAG chatbot using **Firebase** for hosting, **MongoDB** for structured queries, and **PineCone** for vector similarity search
- Integrated **Deepseek V4-flash** as the base LLM; used **pytesseract** for OCR and **whisper** for transcription to extract data from Word, PowerPoint, and PDF files
- Code generation assisted by Agentic AI tools for rapid development

- **RoboMaster Robotics Team**

2022–2023

Mechanical Designer, CAD Engineer

HKUST

- Champion team in the competitive selection process
- Participated in **CAD design** of robotic cars; collaborated with software and hardware teams

- **Robotic Swarms Simulation via Reinforcement Learning**

2025–2026

Research Assistant

Research Project

- Trained a supervised **multi-agent reinforcement learning** model based on zebrafish behavioral video data
- Collaborated with robotics engineering to build a physical robotic fish applying the learned model
- Secured **UROP Research Funding** from HKUST to support the project

ACHIEVEMENTS

- **First Honor Graduate**

- **Champion Team** RoboMaster Team Selection

2022

- **UROP Research Funding** HKUST Undergraduate Research Opportunities Program

2025

- **38th Chinese Physics Olympiad — Bronze Medal** National-level physics competition

2021

TECHNICAL SKILLS

Languages: C++, Python, javascript

Web: React.js, Node.js, tailwind CSS, Firebase

AI/ML: RAG, LLM Integration (Deepseek, Ollama), Reinforcement Learning

Agentic AI: Hermes agent, OpenClaw

Cloud & Data: Google Cloud Platform, VertexAI, MongoDB, PineCone, SQL

Tools: Git, CAD (SolidWorks)